

FIELD THEORY PART–II

Phase 0: Absolute Zero Orientation

From No Idea to Conceptual Starting Point

Academic Article Style Notes for Semester Preparation

Purpose of Phase 0

This phase is written for a student who feels:

“I do not know anything about Field Theory Part–II.”

The aim of this phase is not to solve difficult derivations immediately. The aim is to build the mental picture of the subject.

After this phase, the student should understand:

- what Field Theory Part–II is about,
- why vectors are used,
- why curl and divergence come,
- why magnetic field is connected with current,
- why changing electric field and changing magnetic field are important,
- why electromagnetic waves are produced,
- what kind of answers will be required in semester exam.

One-Sentence Meaning of Part–II

Field Theory Part–II is mainly the study of how magnetic fields are produced by currents, how changing electric and magnetic fields generate each other, and how this mutual generation produces electromagnetic waves.

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1 Absolute Beginning: What Is Field Theory?

Basic Meaning

A field is a quantity that has a value at every point in space.

For example:

- Temperature in a room is a field.
- Air pressure in the atmosphere is a field.
- Electric field around a charge is a field.
- Magnetic field around a current-carrying wire is a field.

In Field Theory, we do not study only one point. We study how a quantity changes from point to point in space.

Simple Example

Suppose there is a hot object in a room. The temperature is high near the object and lower far away.

So, temperature depends on position.

We can write:

$$T = T(x, y, z)$$

This means temperature is a function of position.

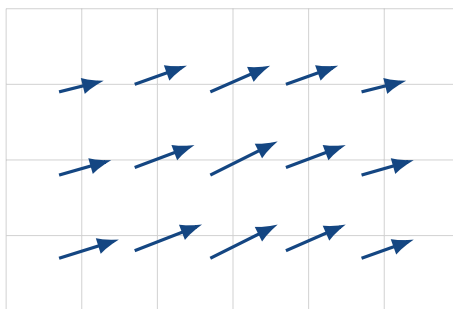
Similarly, electric field may be written as:

$$\mathbf{E} = \mathbf{E}(x, y, z, t)$$

This means electric field can depend on:

$$x, \quad y, \quad z, \quad t$$

So the field may vary in space and time.



A vector field has direction and magnitude at many points

Figure 1: A field can vary from point to point in space.

Phase 0 Learning Point

Field Theory means studying quantities that exist throughout space and may change with position and time.

2 Why Are Vectors Needed?

Scalar and Vector

A scalar has only magnitude.

Examples:

temperature, mass, charge, potential

A vector has magnitude and direction.

Examples:

force, velocity, electric field, magnetic field

Why Field Theory Uses Vectors

Electric and magnetic fields do not only have strength. They also have direction.

For example, if a positive test charge is placed in an electric field, the direction of force on it depends on the direction of the electric field.

So electric field is written as a vector:

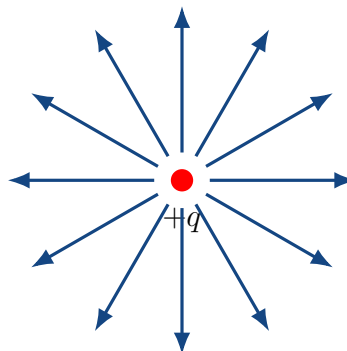
E

Magnetic flux density is also a vector:

B

Magnetic field intensity is also a vector:

H



Electric field has direction at every point

Figure 2: Electric field around a positive point charge is a vector field.

Exam-Oriented Point

Whenever the quantity has direction, write it in bold or with arrow notation:

E, D, B, H, J

3 Main Quantities of Field Theory Part–II

List of Main Quantities

Before starting derivations, memorise the main field quantities.

E, D, H, B, J, S

Symbol	Name	Simple Meaning	Unit
E	Electric field intensity	Force experienced by unit positive charge	V/m
D	Electric flux density	Electric flux per unit area	C/m^2
H	Magnetic field intensity	Magnetising field produced by current	A/m
B	Magnetic flux density	Magnetic flux per unit area	Tesla (T)
J	Current density	Current flowing per unit area	A/m^2
S	Poynting vector	Electromagnetic power flow per unit area	W/m^2
ρ_v	Volume charge density	Charge per unit volume	C/m^3
ϵ	Permittivity	Ability of medium to support electric field	F/m
μ	Permeability	Ability of medium to support magnetic field	H/m
σ	Conductivity	Ability of medium to conduct current	S/m

Most Important Relation Set

$$\mathbf{D} = \epsilon \mathbf{E}$$

$$\mathbf{B} = \mu \mathbf{H}$$

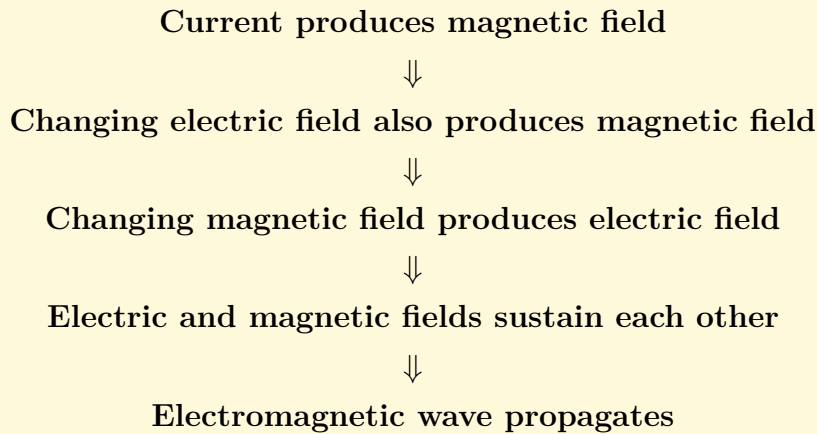
$$\mathbf{J} = \sigma \mathbf{E}$$

These three relations are used again and again in Maxwell equations and EM wave derivation.

4 The Story of Part–II in Simple Words

Main Story

The whole Part–II can be understood as one story.



Block-by-Block Meaning

Block 1: Current produces magnetic field

A steady current produces a magnetic field around it.

This is described by:

$$\nabla \times \mathbf{H} = \mathbf{J}$$

Block 2: Changing electric field also produces magnetic field

Maxwell discovered that a changing electric flux density acts like current.

This gives:

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

Block 3: Changing magnetic field produces electric field

A time-varying magnetic field produces a circulating electric field.

This gives:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Block 4: Together they create wave

A changing electric field creates magnetic field. A changing magnetic field creates electric field.

This self-sustaining process travels as electromagnetic wave.

5 How Current Produces Magnetic Field

Magnetostatic Beginning

Magnetostatics deals with steady current and steady magnetic field.

The word can be split as:

magneto + static

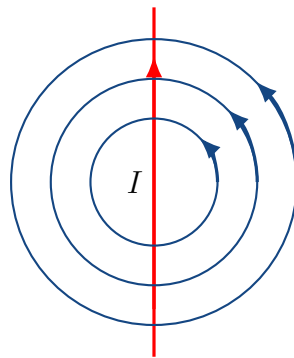
It means magnetic field that does not change with time.

Simple Wire Example

When current flows through a straight conductor, magnetic field lines form closed circles around the conductor.

The direction is given by the right-hand thumb rule:

- thumb points in direction of current,
- curled fingers show direction of magnetic field.



Current produces circular magnetic field lines

Figure 3: Magnetic field around a current-carrying wire.

Important Exam Link

This physical idea becomes the mathematical equation:

$$\nabla \times \mathbf{H} = \mathbf{J}$$

Meaning:

Current density creates circulation of magnetic field intensity.

6 Why Curl Comes in This Subject

Simple Meaning of Curl

Curl means local circulation.

If a vector field has a tendency to rotate around a point, then curl is present.

If there is no rotational tendency, curl is zero.

Paddle-Wheel Picture

Imagine placing a small paddle wheel inside a flowing field.

If the paddle wheel rotates, the curl is non-zero.

If the paddle wheel does not rotate, the curl may be zero.

In electromagnetic field theory:

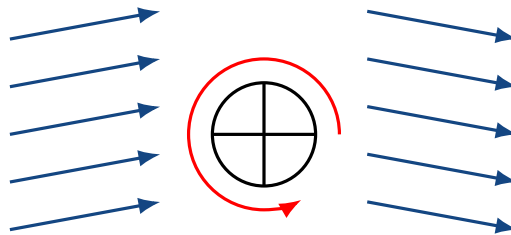
$$\nabla \times \mathbf{H}$$

means circulation tendency of magnetic field intensity.

$$\nabla \times \mathbf{E}$$

means circulation tendency of electric field intensity.

Curl means local rotation or circulation



If the paddle rotates, curl is present

Figure 4: Physical picture of curl using a paddle wheel.

Phase 0 Memory

Curl is used because electromagnetic fields often circulate:

Current causes circulating $\mathbf{H}$

Changing $\mathbf{B}$ causes circulating $\mathbf{E}$

7 Why Divergence Comes in This Subject

Simple Meaning of Divergence

Divergence means source or sink strength.

If field lines come out from a point, divergence is positive.

If field lines go into a point, divergence is negative.

If no field line starts or ends there, divergence is zero.

Field-Line Meaning

For electric field:

$$\nabla \cdot \mathbf{D} = \rho_v$$

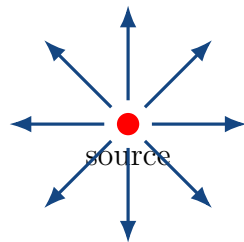
This means charge is the source of electric flux density.

For magnetic field:

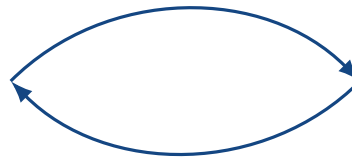
$$\nabla \cdot \mathbf{B} = 0$$

This means magnetic field has no source or sink.

So magnetic field lines are closed loops.



positive divergence



zero divergence for closed loops

Figure 5: Divergence tells whether field lines start or end.

Important Part-II Result

For magnetic field:

$$\nabla \cdot \mathbf{B} = 0$$

Meaning:

Magnetic field lines are closed and no isolated magnetic pole exists.

8 Why Stokes' Theorem Comes

Need of Stokes' Theorem

Stokes' theorem connects circulation around a boundary with curl over the surface. It is one of the most important mathematical tools in Part-II.

Statement

Stokes' theorem is:

$$\oint_C \mathbf{A} \cdot d\mathbf{l} = \iint_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$

Left side:

$$\oint_C \mathbf{A} \cdot d\mathbf{l}$$

means circulation around the boundary.

Right side:

$$\iint_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$

means total curl passing through the surface.

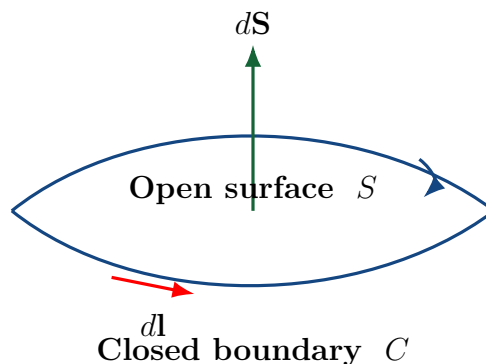


Figure 6: Stokes' theorem: boundary circulation equals surface integral of curl.

Why It Is Important

Stokes' theorem is used to convert:

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = I$$

into:

$$\nabla \times \mathbf{H} = \mathbf{J}$$

So without Stokes' theorem, Ampere's differential law cannot be established clearly.

9 The First Core Equation: Ampere's Law

Basic Meaning

Ampere's law says:

Current produces magnetic field circulation.

In integral form:

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}}$$

In differential form:

$$\nabla \times \mathbf{H} = \mathbf{J}$$

How to Think

When current passes through a wire, magnetic field circulates around it.
This circulation is measured by curl:

$$\nabla \times \mathbf{H}$$

The cause of this circulation is current density:

$$\mathbf{J}$$

So:

$$\nabla \times \mathbf{H} = \mathbf{J}$$

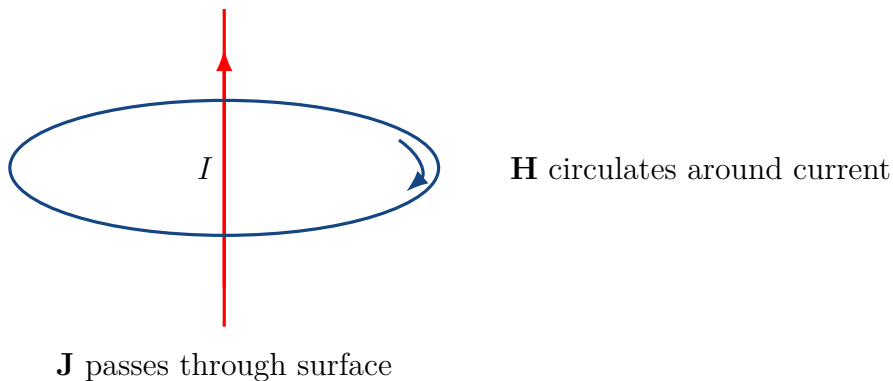


Figure 7: Current density creates magnetic field circulation.

Semester Writing Point

Whenever the question asks:

$$\text{Establish } \nabla \times \mathbf{H} = \mathbf{J}$$

write:

1. Ampere's circuital law,

2. current density surface integral,
3. Stokes' theorem,
4. compare both surface integrals,
5. final boxed equation.

10 Why Ampere's Law Needed Modification

Problem

The old Ampere law:

$$\nabla \times \mathbf{H} = \mathbf{J}$$

works properly for steady current.

But in time-varying situations, it becomes incomplete.

Capacitor Example

Consider a capacitor connected to an AC source.

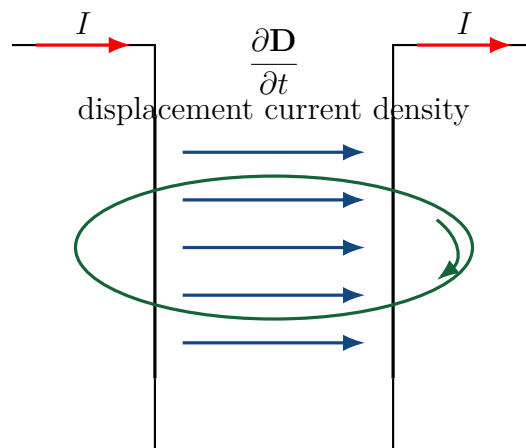
In the wire, there is conduction current.

Between capacitor plates, there is no actual conduction current because the dielectric gap blocks charge flow.

But still magnetic field exists around the capacitor region.

This means something equivalent to current exists between the plates.

Maxwell called it displacement current.



Changing electric field between plates acts like current

Figure 8: Displacement current between capacitor plates.

Modified Ampere Law

Maxwell added displacement current density:

$$\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t}$$

So the complete law becomes:

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

11 The Second Core Equation: Faraday's Law

Basic Meaning

Faraday's law says:

Changing magnetic field produces circulating electric field.

Differential form:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Simple Explanation

If magnetic field changes with time, then an electric field is induced.

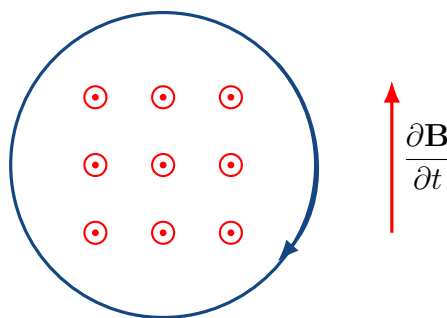
This induced electric field is not like electrostatic field from a stationary charge.

It forms closed loops.

Therefore it has curl.

The negative sign represents Lenz's law.

Changing magnetic field



Induced circulating electric field $\mathbf{E}$

Figure 9: Changing magnetic field creates circulating electric field.

Semester Writing Point

When asked to derive:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

write:

1. Faraday law in integral form,
2. magnetic flux definition,
3. apply Stokes' theorem,
4. compare surface integrals,
5. explain negative sign as Lenz's law.

12 The Complete Maxwell Picture

Why Maxwell Equations Matter

Maxwell equations are the final compact form of electromagnetic field theory.

For Part-II, the most important ones are:

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

and

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

These two equations explain electromagnetic waves.

Four Main Maxwell Equations

In differential form:

$$\nabla \cdot \mathbf{D} = \rho_v$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

No.	Equation	Meaning	Exam Use
1	$\nabla \cdot \mathbf{D} = \rho_v$	Electric charge is source of electric flux density.	Background concept
2	$\nabla \cdot \mathbf{B} = 0$	Magnetic flux has no source or sink.	Short note
3	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	Changing magnetic field creates circulating electric field.	Long answer
4	$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$	Current and changing electric field create circulating magnetic field.	Long answer

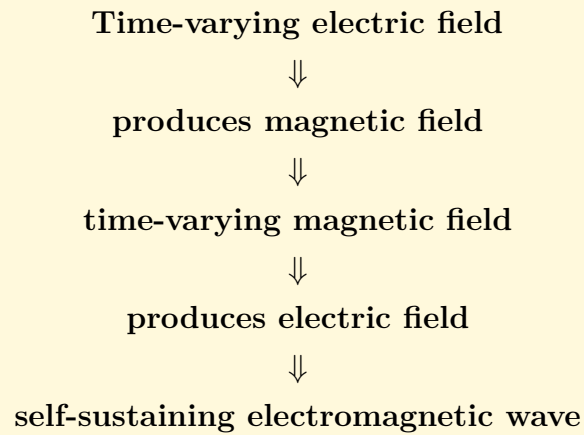
Phase 0 Main Idea

Maxwell equations show that electric and magnetic fields are not separate. They are connected parts of one electromagnetic field.

13 How Electromagnetic Waves Are Born

The Main Chain

Electromagnetic wave formation is the most important final concept of Part-II.
The logic is:



Simple Explanation

An electromagnetic wave does not need a material medium like sound.

It can travel in free space.

In free space:

$$\sigma = 0$$

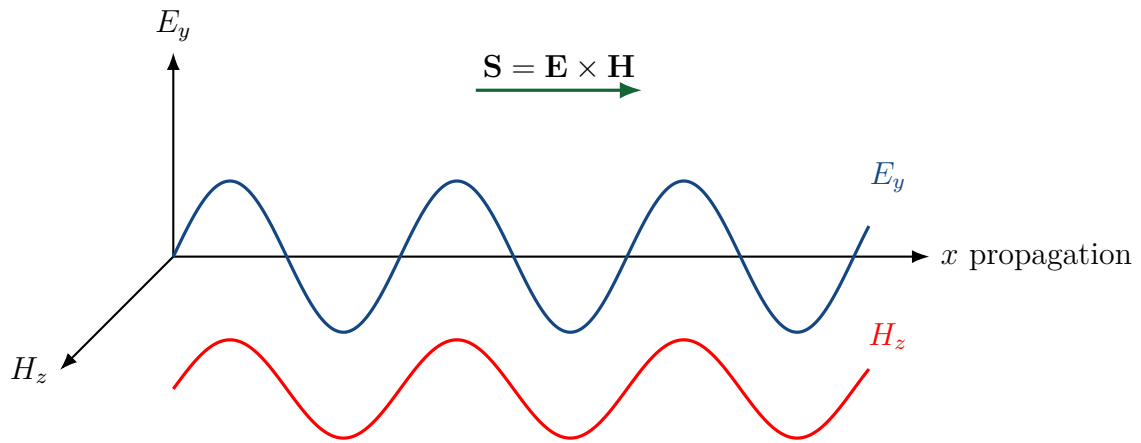
and

$$\mu = \mu_0, \quad \epsilon = \epsilon_0.$$

The wave velocity is:

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

This is the speed of light.



Plane EM wave: $\mathbf{E}$, $\mathbf{H}$ and propagation direction are mutually perpendicular

Figure 10: Basic picture of electromagnetic wave propagation.

Exam Sentence

An electromagnetic wave is produced by the mutual regeneration of time-varying electric and magnetic fields.

14 What Is a Plane Electromagnetic Wave?

Definition

A plane electromagnetic wave is a wave in which the electric and magnetic field values are constant over any plane perpendicular to the direction of propagation.

Simple Picture

If the wave travels in the $+x$ direction, then a plane perpendicular to x is the yz -plane.
At a fixed value of x , the field is same everywhere on the yz -plane.
So the field varies only with:

$$x$$

and time:

$$t$$

Standard Assumption

For a plane wave travelling along $+x$ direction:

$$\frac{\partial}{\partial y} = 0$$

$$\frac{\partial}{\partial z} = 0$$

Only variation with x remains.

Common Field Directions

A standard assumption is:

$$\mathbf{E} = E_y(x, t)\hat{\mathbf{y}}$$

$$\mathbf{H} = H_z(x, t)\hat{\mathbf{z}}$$

Propagation is along:

$$\mathbf{E} \times \mathbf{H}$$

Since:

$$\hat{\mathbf{y}} \times \hat{\mathbf{z}} = \hat{\mathbf{x}}$$

the wave travels along $+x$.

15 What Is Wave Equation?

Basic Meaning

A wave equation is a differential equation whose solution represents a travelling wave.

Standard Form

The one-dimensional wave equation is:

$$\frac{\partial^2 \psi}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2}$$

where:

$$\psi$$

is the wave quantity and

$$v$$

is wave velocity.

EM Wave Form

For electric field in free space:

$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

For magnetic field in free space:

$$\nabla^2 \mathbf{H} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{H}}{\partial t^2}$$

Comparing with the standard wave equation:

$$\frac{1}{v^2} = \mu_0 \epsilon_0$$

So:

$$v = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

Phase 0 Memory

Whenever you see:

$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

think:

This is the wave equation of electric field in free space.

16 What Is Poynting Vector?

Basic Meaning

Poynting vector tells the direction and amount of electromagnetic energy flow.

Definition

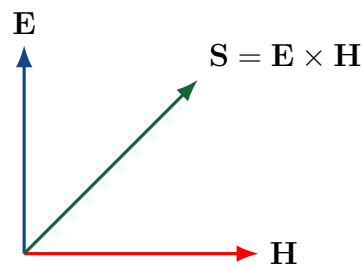
Poynting vector is:

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}$$

Its unit is:

$$W/m^2$$

It tells power flow per unit area.



Direction of EM energy flow

Figure 11: Poynting vector direction.

Exam Use

Poynting vector appears in:

- short notes,
- Poynting theorem,
- EM wave propagation,
- energy flow explanation.

17 What Is Polarization?

Basic Meaning

Polarization describes how the electric field vector of an electromagnetic wave varies with time at a fixed point.

Why Electric Field Is Used

In electromagnetic waves, both $\mathbf{E}$ and $\mathbf{H}$ exist.
But polarization is defined mainly by the direction and variation of the electric field vector $\mathbf{E}$.
So when asked about polarization, focus on $\mathbf{E}$.

Path traced by the tip of electric field vector

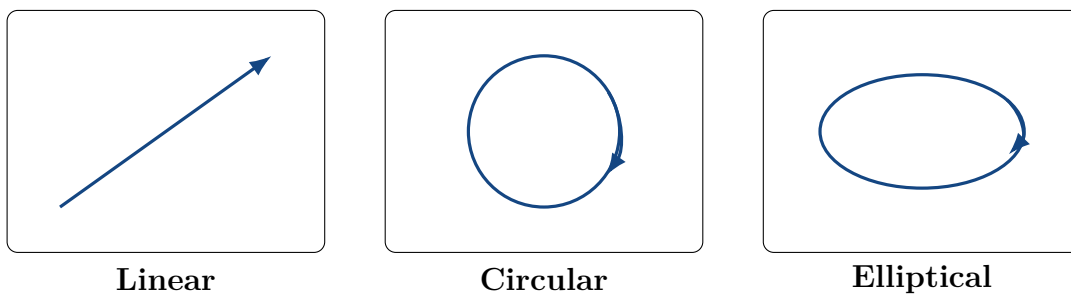


Figure 12: Three basic types of electromagnetic wave polarization.

Three Types

Linear polarization

Circular polarization

Elliptical polarization

18 What Is Magnetic Boundary Condition?

Basic Meaning

When magnetic field passes from one medium to another, the field changes direction and magnitude according to boundary conditions.

Two Main Conditions

At a boundary between two magnetic media:

$$H_{1t} = H_{2t}$$

if there is no surface current.

Also:

$$B_{1n} = B_{2n}$$

These are the tangential and normal component conditions.

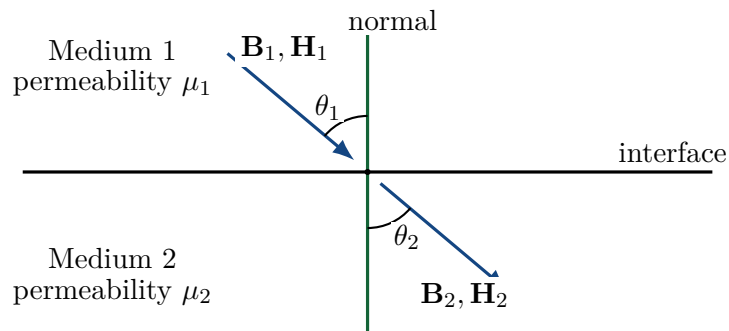


Figure 13: Magnetic field passing through two magnetic media.

Boundary Relation

Using:

$$\mathbf{B}_1 = \mu_1 \mathbf{H}_1$$

and

$$\mathbf{B}_2 = \mu_2 \mathbf{H}_2$$

we get:

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{\mu_1}{\mu_2}$$

19 How to Recognise Part–II Questions

Question Pattern Recognition

Part–II questions can be recognised by keywords.

If you identify the keyword, you can decide the answer pattern quickly.

No.	Keyword in Question	Topic	Answer Pattern
1	Definition of curl	Curl derivation	Definition, small loop, formula, significance
2	Physical significance of curl	Curl meaning	Paddle wheel, rotation, right-hand rule
3	Stokes' theorem	Vector theorem	Statement, surface diagram, cancellation idea
4	$\nabla \times \mathbf{H} = \mathbf{J}$	Ampere law	Ampere integral form, Stokes theorem, compare
5	Boundary condition	Magnetic media	Interface diagram, H_t , B_n , relation
6	Displacement current	Modified Ampere law	Capacitor diagram, $J_d = \partial D / \partial t$
7	$\nabla \times \mathbf{E}$	Faraday law	Integral form, Stokes theorem, Lenz sign
8	EM wave equation	Maxwell wave derivation	Curl equations, curl of curl identity, final equation
9	Plane wave	EM wave solution	Assume direction, solve sinusoidal form
10	Poynting	Energy flow	$S = E \times H$, theorem, unit, meaning
11	Polarization	EM wave property	Linear, circular, elliptical
12	$\nabla \cdot B = 0$	Magnetic divergence	No monopole, closed field lines
13	$\nabla \cdot J = 0$	Steady current	No charge accumulation, KCL meaning
14	Biot-Savart	Magnetic field due to current	Formula, diagram, integration/numerical

20 Phase 0 Mini Glossary

Why This Glossary Is Needed

Before studying derivations, the symbols must not look frightening.

This section gives beginner-friendly meanings.

No.	Symbol	Simple Meaning	Where It Appears
1	∇	Vector differential operator	Gradient, divergence, curl
2	$\nabla \cdot$	Divergence operator	Source/sink measurement
3	$\nabla \times$	Curl operator	Circulation measurement
4	$\mathbf{E}$	Electric field intensity	Faraday law, wave equation
5	$\mathbf{D}$	Electric flux density	Modified Ampere law
6	$\mathbf{H}$	Magnetic field intensity	Ampere law, wave equation
7	$\mathbf{B}$	Magnetic flux density	Faraday law, divergence law
8	$\mathbf{J}$	Current density	Ampere law
9	$\mathbf{S}$	Poynting vector	Energy flow
10	ρ_v	Volume charge density	Continuity equation
11	μ	Permeability	Magnetic medium
12	ϵ	Permittivity	Electric medium
13	σ	Conductivity	Conducting medium wave equation
14	ω	Angular frequency	Plane wave solution
15	β	Phase constant	Plane wave solution
16	c	Speed of light in free space	EM wave velocity

21 Phase 0 Formula Preview

Purpose

Do not memorise all formulas now. Just become familiar with them.

Preview List

$$\nabla \times \mathbf{H} = \mathbf{J}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \cdot \mathbf{J} = 0$$

$$\mathbf{B} = \mu \mathbf{H}$$

$$\mathbf{D} = \epsilon \mathbf{E}$$

$$\mathbf{J} = \sigma \mathbf{E}$$

$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

$$\nabla^2 \mathbf{H} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{H}}{\partial t^2}$$

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}$$

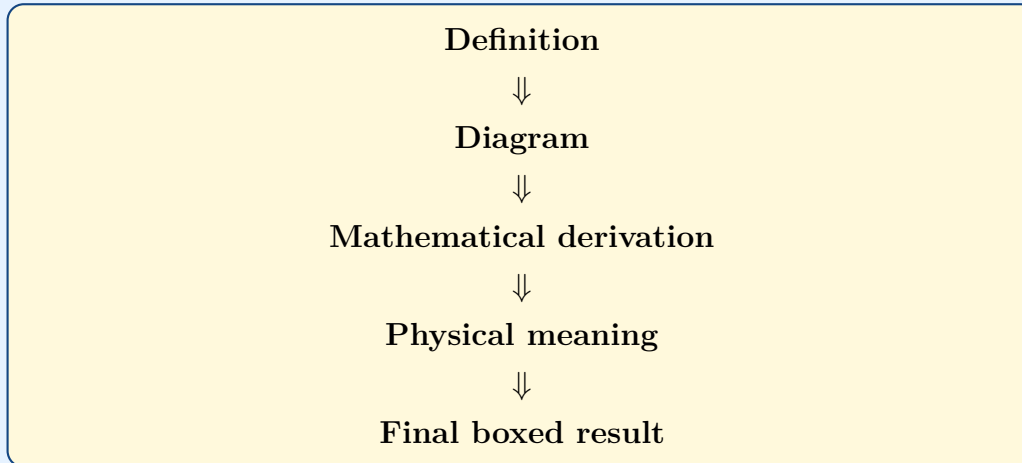
$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

22 What a Student Should Write in Semester Answers

Semester Answer Style

For Field Theory Part–II, answers should not be formula-only.

Use this structure:



For Long Answers

For long answers like curl, EM wave equation, Poynting theorem, magnetic boundary condition, write:

1. one short introduction,
2. one neat labelled diagram,
3. step-by-step derivation,
4. final formula in a box,
5. physical interpretation.

For Short Notes

For short notes like Poynting vector, displacement current, polarization, write:

1. definition,
2. formula,
3. unit if required,
4. physical meaning,
5. one application or diagram if possible.

Golden Rule

In Field Theory, a neat diagram and clear final boxed formula can improve the answer quality strongly.

23 Phase 0 Self-Test

Instructions

Before going to Phase 1, try to answer these orally.
No full derivation is required yet.

1. What is a field?
2. Why is electric field a vector?
3. What is the meaning of $\mathbf{E}$?
4. What is the meaning of $\mathbf{H}$?
5. What is the meaning of $\mathbf{B}$?
6. What is the meaning of $\mathbf{D}$?
7. What is current density $\mathbf{J}$?
8. What does curl roughly mean?
9. What does divergence roughly mean?
10. Why does magnetic field have zero divergence?
11. What does Stokes' theorem connect?
12. What does Ampere's law say physically?
13. Why did Maxwell add displacement current?
14. What does Faraday's law say physically?
15. How are EM waves created?
16. What is a plane wave?
17. What does Poynting vector represent?
18. What is polarization?
19. What are linear, circular and elliptical polarization?
20. What is the speed of EM wave in free space?

Pass Condition

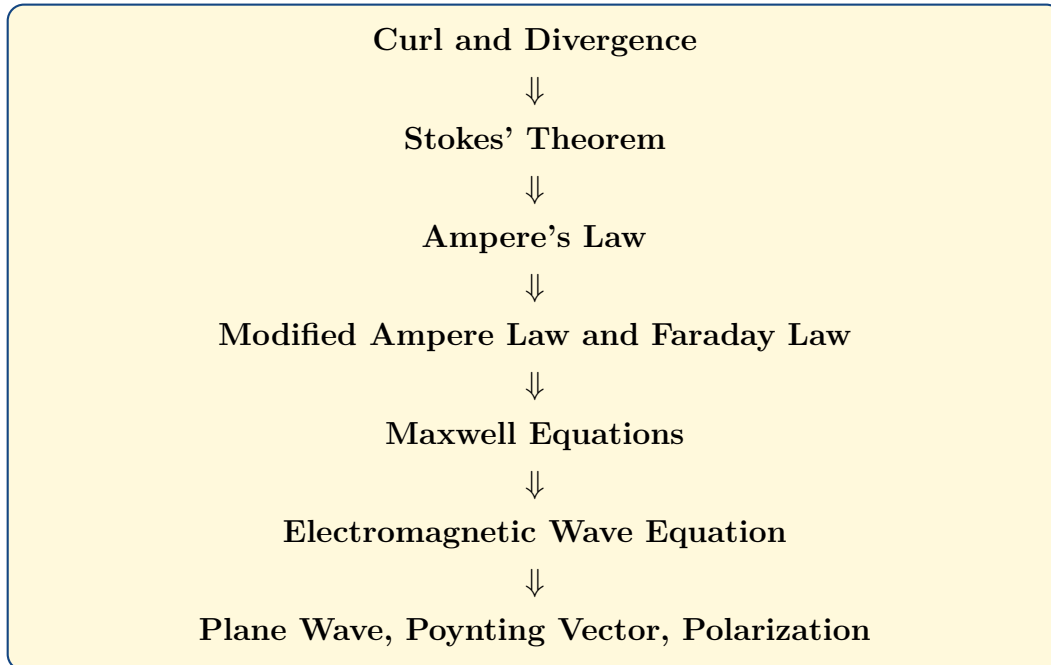
If you can explain at least 15 out of these 20 questions in simple words, Phase 0 is complete.

24 Phase 0 Final Revision Sheet

One-Page Summary

Field Theory Part-II begins with vector calculus and ends with electromagnetic waves.

Core Chain



Final Phase 0 Conclusion

Phase 0 is complete when the student understands the basic story:

Current and changing fields create electromagnetic effects.

Maxwell equations combine those effects.

Electromagnetic waves are the final result of time-varying electric and magnetic fields.

Now the student is ready to start Phase 1: Basic Vector Language and Coordinate System.

25 Transition from Phase 0 to Phase 1

What Comes Next

Phase 1 will not jump directly into Maxwell equations.
It will first build the mathematical language:

vectors $\rightarrow$ unit vectors $\rightarrow$ differential length $\rightarrow$ differential surface $\rightarrow$ line integral $\rightarrow$ surface integral

Why This Transition Is Necessary

Most students feel Field Theory is difficult because they try to memorise formulas without understanding the notation.

For example:

$$\oint_C \mathbf{H} \cdot d\mathbf{l}$$

looks difficult only until one understands:

$$\mathbf{H}$$

is a vector field,

$$d\mathbf{l}$$

is a small length vector,
and

$$\oint_C$$

means adding around a closed path.

So Phase 1 will make the notation friendly.

Final Instruction Before Phase 1

Do not worry about memorising derivations yet.

Only keep this mental map:

Part-II is about circulation, source/sink, current, changing fields and waves.